

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations. Whether decomposing this heterogeneous task into specialized agents improves reliability over an equivalent single agent remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system assigning queries to four bounded specialists with scoped tools and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid retrieval with neural reranking for regulations. Scenarios 1 and 2 use a 100-query held-out benchmark for architecture comparison and ablation, respectively. The former compares CTU-Chat with a matched single-agent baseline and a routed generic variant; the latter removes specialist policies, tool isolation, heterogeneous evidence, deterministic calculation, and route repair. Scenario 3 uses 50 separate cases to stress-test tool ambiguity across 11, 31, and 51 tools. In architecture comparison, CTU-Chat achieves a 10.0 pp higher Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$) point estimate (76.0% vs. 66.0%; paired 95% CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$) with 41.4% fewer recorded specialist input tokens, alongside an unlogged upstream routing overhead of ~ 650 tokens and sequential dispatch latency (10.65 s pipeline vs. 5.16 s post-retrieval baseline). Under registry expansion to 51 tools, the scoped specialist achieves 88.0% tool E2E Success (vs. 78.7% for Top-10 selection). These findings suggest that domain-scoped specialization and heterogeneous evidence allocation may improve fact-pass performance under the evaluated conditions, with measurable trade-offs in coordination latency and token consumption.

Keywords: Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

At the opening of the 2025–2026 academic year, Can Tho University (CTU) reported 48,960 undergraduate students¹. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [8] and REBot for regulation advising [7]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [6,3,2], while agentic architectures support tool use and multi-agent coordination [18,16,14]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [12] further motivate deterministic calculation for financial queries.

Central Hypothesis. When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with specialized evidence paths can improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges highlight three primary research gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three primary contributions: (1) **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; (2) **C2**—an architecture-level comparative evaluation against a matched single-agent baseline and a routed generic variant; and (3) **C3**—architectural ablations of five components and an 11–51 tool ambiguity stress test.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental

¹ CTU opening ceremony report, 5 September 2025: <https://en.ctu.edu.vn/news-t/can-tho-university-inaugurates-the-2025-2026-academic-year-embarking-on-a-new-era-and-affirming-its-role-as-the-leading-institution-in-education-training-science-and-technology-in-the-mekong-delta.html> (accessed 28 September 2026).

setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [9]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, which necessitates the coupling of dense embeddings with exact lexical matching.

BM25 [13] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [5,6] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [1] merges BM25 and dense candidates without score calibration, while neural cross-encoders [10,17] jointly encode query–passage pairs for reranking. Adaptive-RAG [4] demonstrates complexity-based retrieval selection. These approaches support dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [11]. GraphRAG [2] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [18] demonstrates interleaved reasoning and action, AutoGen [16] provides a multi-agent conversation framework, and Toolformer [14] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [15]. CAAS supports admission counseling for IT programs [8], while REBot combines retrieval with category-guided GraphRAG for academic regulations [7]. CTU-Chat differs from prior systems

in three aspects: (1) *Task scope*: representing 111 distinct program codes from 113 curriculum files in relational property graphs, with deterministic tuition arithmetic; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component ablations.

2.5 Research Gaps

An analysis of the reviewed literature reveals three primary gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

3 Proposed Model

CTU-Chat combines three design principles: centralized coordination, heterogeneous knowledge allocation, and representation-aware evidence routing. A contract-checked supervisor assigns one specialist per query. Depending on the domain, the specialist uses graph traversal for curricula, structured lookup and calculation for fees, or retrieval for narrative policies.

3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-Chat uses a typed `AgentState` for rewrites, routes, evidence, and responses. An intent-owner contract checks supervisor output before conditional dispatch: academic queries enter the graph agent; other domains use intent-scoped retrieval. Specialists see only assigned tools and return to END without peer exchanges. The fixed-query benchmark in Section 4 bypasses Redis history and query rewriting used in live multi-turn serving.

3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph is built from 113 curriculum Markdown files representing 111 distinct program codes. The resulting graph stores approximately 7,000 nodes across 10 labels: curriculum entities (e.g., `Program`, `Course`, `Block`, `PLO`, `PEO`, `Track`, `Job`) and tuition entities (`TuitionFee`, `TuitionPolicy`, `ExemptionRate`) connected by over 16,000 edges across 13 typed relationships encoding containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names,

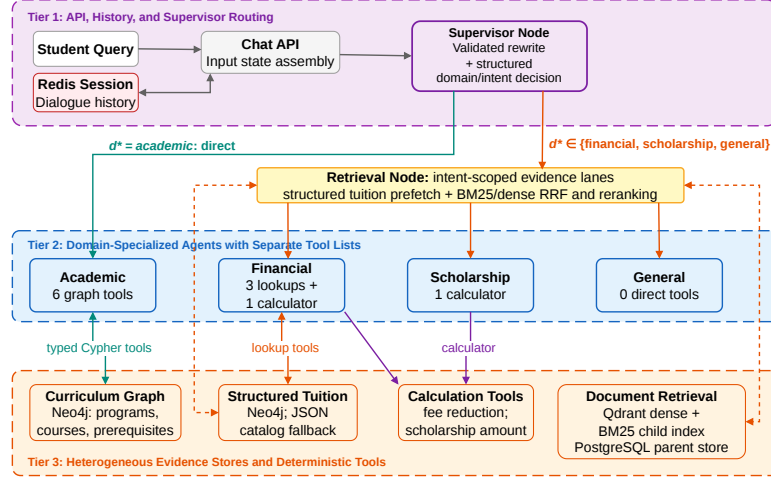


Fig. 1. Implemented CTU-Chat architecture. Redis-backed history and validated query rewriting are used in live multi-turn serving; the offline benchmark uses fixed single-turn queries and bypasses these components. The supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

3.3 Routing and Deterministic Arithmetic

For a query q and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain d_{raw} and intent i_{raw} under a structured schema, falling back to **general/other** on parsing failure. The ownership contract resolves rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

Deterministic Tuition Arithmetic. For fee-reduction requests, the tool `tinh_toan_hoc_phi` receives three parameters—per-credit tuition T , per-credit exemption basis B , and exemption percentage $r \in [0, 100]$ —and calculates deductible credit amount $D = B \times (r/100)$ alongside net payable tuition $T_{\text{net}} = \max(0, T - D)$. The tool clamps T_{net} at zero. The primary contribution is the integration of validated rewriting, constrained routing, and deterministic arithmetic into a unified multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

CTU-Chat stores curricula in Neo4j, accesses cohort-specific fees through structured lookup, and retrieves narrative regulations from documents. Financial requests try Neo4j tuition lookup with JSON fallback before the repaired intent selects metadata-filtered retrieval lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF: $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$. The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

3.5 Cross-Domain Query Handling

For cross-domain queries, CTU-Chat uses a single-owner, signal-based evidence-prefetch policy. The supervisor selects an owner by dominant intent; keyword signals add academic-program, academic-regulation, tuition, or scholarship retrieval lanes. The owner uses this evidence without calling another specialist. This bounds execution but may leave a needed tool or source outside the owner’s scope. The 20 cross-domain held-out queries test this policy.

4 Experimental Setup

4.1 Dataset and Annotation

We evaluate CTU-Chat on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 38 single-source, 62 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

4.2 Baselines and Configurations

Baseline equivalence and execution bounds. Scenario 1 configurations share Gemini 2.5 Flash Lite ($T=0.0$), knowledge representations, deterministic calculator access, and the same retrieval output (BM25 + dense + RRF + `bge-reranker-v2-m3`, $k=7$).

Shared retrieval gives identical Source Recall (78.6%) and Source AP (70.8%); tool outputs remain configuration-specific. Scenario 1 compares orchestration bundles, while Scenario 2 changes one component at a time. Multi-agent variants allow at most 3 tool calls, 2,048 output tokens, and 30-second timeouts. Reported tokens cover specialist execution; supervisor routing adds an estimated ~650 input tokens per query.

Scenario 1 — Architecture Comparison (RQ1): Three configurations are evaluated on the held-out set ($N=100$ queries, 300 runs under greedy decoding $T=0.0$): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

Scenario 2 — Ablation (RQ2): Five ablations each remove one component from Full CTU-Chat ($N=100$ queries per ablation, $T=0.0$): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

Scenario 3 — Tool Ambiguity (RQ3): Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding $50 \times 4 \times 3 \times 3 = 1,800$ decisions across counterbalanced blocks.

4.3 Evaluation Metrics

Reported metrics span four evaluation dimensions: (1) *Final-answer quality:* Required-Fact Coverage (FactCov, continuous proportion of reference facts matched via substring containment in generated responses) and Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$). (2) *Evidence retrieval:* Source Recall (Src.R) and Source AP (Src.AP) assess Scenario 1. Table 4 ranks unique source filenames after collapsing repeated chunks. Per query, $\text{Hit}@k$: any gold in top k ; $\text{P}@5$: top-five gold count divided by five; $\text{R}@5$: that count divided by gold count (1 if none); $\text{MRR}@10$: reciprocal first gold rank within ten (0 if absent). Means are over 100 queries. (3) *Execution reliability:* Tool Invocations per Query, Tool Selection Accuracy, Tool End-to-End Success (E2E), and route repair rate. (4) *Operational cost:* Specialist input/output tokens and wall-clock latency.

Primary Evaluation Metrics. In Scenarios 1 and 2, the primary evaluation metric is the *Automated Fact Pass Rate* ($\text{Pass}_{\geq 50\%}$), defined programmatically as the proportion of responses where at least 50% of predefined required fact phrases are matched via automated substring containment ($\text{FactCov} \geq 0.50$), complemented by continuous Fact Coverage. Since this lexical metric measures surface phrase occurrence rather than semantic or arithmetic accuracy, it functions as a surface-grounding proxy. To address this limitation, it is complemented in Scenario 3 by deterministic *Tool E2E Success* (correct tool selection, valid parameters, and error-free execution).

4.4 Statistical Protocol and Measurement Scope

Evaluations use greedy decoding ($T=0.0$) to minimize sampling variance. Paired differences across configurations are computed at the query level ($N=100$); 95% CIs are estimated using 10,000 bootstrap resamples. The primary confirmatory hypothesis compares CTU-Chat versus Unified Single Agent on $\text{Pass}_{\geq 50\%}$. Timing follows configuration boundaries: baseline S1-A latency measures agent invocation post-retrieval, while multi-agent variants measure pipeline latency inclusive of supervisor routing, route repair, and specialist execution. Reported tokens record specialist agent execution; upstream supervisor routing adds an estimated overhead of ~ 650 input tokens per query (derived from prompt and schema length).

4.5 Infrastructure

We use Gemini 2.5 Flash Lite via Vertex AI ($T=0.0$) and rerank on a GTX 1650. Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800 characters, 100 overlap); Qdrant indexes child chunks (400 characters, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. The system uses bge-reranker-v2-m3, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

5 Results

5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares three arms on the 100-query held-out set. Shared pre-generation retrieval fixes Source Recall (78.6%) and Source AP (70.8%) across arms; answer and cost differences reflect their combined orchestration changes.

Table 1. Scenario 1: Architecture-level comparison ($N = 100$). Pass_{50} = Automated Fact Pass Rate ($\geq 50\%$ required facts). Tools/Q = mean tool invocations ($\downarrow$). In/Out Tok = specialist agent tokens; estimated routing input (~ 650 tokens/query) is excluded. Bold = best reliability / lowest cost.

Arch.	Pass_{50}	Fact	Src.R	Src.AP	Tools/Q	In Tok	Out Tok	Lat. (ms)
Single Agent	66.0	57.5	78.6	70.8	0.41	6296	1530	5159
Routed Gen.	66.0	54.2	78.6	70.8	0.31	3473	1415	8921
CTU-Chat	76.0	61.0	78.6	70.8	0.28	3690	2092	10652

Pairwise bootstrap analysis (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves a 10.0 pp higher $\text{Pass}_{\geq 50\%}$ point estimate (76.0% vs. 66.0%, paired 95% bootstrap CI $[+2.0, +19.0]$ pp; exact McNemar $p=0.0414$), alongside higher Fact Coverage ($\Delta = +3.58$ pp, CI $[-1.2, +8.5]$ pp) and 41.4% fewer specialist input tokens, under differing measurement boundaries (10,652 ms end-to-end pipeline vs. 5,159 ms

post-retrieval generation). (2) CTU-Chat vs. Routed Generic: CTU-Chat has +6.85 pp higher Fact Coverage (CI [+0.7, +13.0] pp, $p < 0.05$). (3) Routed Generic and Single Agent both pass 66.0%, with Fact Coverage of 54.2% and 57.5%, respectively. The Single Agent invokes more tools (0.41 per query); prompt interference was not measured.

On the 20 cross-domain queries, CTU-Chat achieves an 80.0% pass rate (vs. 70.0% Single, 65.0% Generic). Evaluating sensitivity across thresholds, CTU-Chat demonstrates a higher automated pass rate at the baseline 50% fact-coverage threshold (unadjusted exact McNemar $p = 0.0414$). This margin narrows at 60% (54.0% vs. 50.0%) and reverses at stricter thresholds (e.g., 27.0% vs. 30.0% at 75%). However, the difference at 75% remains statistically uncertain ($p = 0.5488$, 95% CI [-10.0, +3.0] pp), indicating neither architecture holds a definitive advantage under stringent fact-matching criteria.

Operational Trade-offs: Table 1 shows CTU-Chat has the highest automated pass estimate and fewest tool calls per query. Latencies span different measurement boundaries, precluding a direct speed comparison. Among multi-agent variants, CTU-Chat records 1,731 ms and 677 output tokens more than Routed Generic. Its specialist uses 3,690 input tokens versus 6,296 for Single Agent, plus an estimated ~650 routing tokens.

5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ($N=100$). Deltas are relative to the full system.

Table 2. Scenario 2: Architectural ablation results ($N = 100$). Panel A reports overall benchmark differences (Δ relative to Full CTU-Chat). Panel B reports Automated Fact Pass Rate ($\text{Pass}_{\geq 50\%}$, %) across query families.

Panel A: Overall Benchmark Ablations					
Ablated Component	$\Delta\text{Pass}_{.50}$	ΔFact	$\Delta\text{Src.R}$	ΔTok	ΔLat
No Spec. Policy	-10.0	-6.85	0.0	-217	-1731
Full Tool Vis.	0.0	+0.93	0.0	+1270	+296
Uniform Text-RAG	-9.0	-2.18	-12.3	-300	-1910
No Route Rep.	-2.0	-1.27	0.0	-219	-1610
No Determ. Calc. [†]	-4.0	-0.93	0.0	-782	-4540
Panel B: Fact-Grounded Pass Rate ($\text{Pass}_{\geq 50\%}$) by Query Family					
Configuration	Graph ($n=19$)	Struct-fin ($n=28$)	Narr-reg ($n=33$)	Cross ($n=20$)	
Full CTU-Chat	63.2	75.0	81.8	80.0	
No Spec. Policy	52.6	67.9	72.7	65.0	
Full Tool Vis.	68.4	67.9	84.8	80.0	
Uniform Text-RAG	63.2	60.7	78.8	60.0	
No Route Rep.	68.4	78.6	78.8	65.0	

[†]Full benchmark with calculators disabled. $\Delta\text{Tok}/\Delta\text{Lat}$ in tokens/ms per query.

The largest pass-rate decreases occur without Specialist Policy (-10.0 pp, 95% CI [-19.0, -1.0] pp) and with Uniform Text-RAG (-9.0 pp, CI [-15.0, -4.0] pp). The

former lowers Fact Coverage by 6.85 pp (CI $[-13.0, -0.7]$ pp); the latter lowers Source Recall by 12.3 pp (CI $[-16.5, -8.3]$ pp). Removing tool isolation leaves the pass estimate unchanged (CI $[-6.0, +6.0]$ pp) but adds 1,270 input tokens. Without route repair, overall pass changes by -2.0 pp (CI $[-7.0, +3.0]$ pp), while cross-domain pass falls from 80.0% to 65.0%. Without deterministic calculation, overall automated fact pass changes by -4.0 pp (95% CI $[-9.0, 0.0]$ pp). This benchmark-wide result uses the FactCov ≥ 0.50 threshold and does not measure numeric exact match. Panel B shows 63.2% pass on graph queries versus 80.0% on cross-domain queries. Graph queries can require credit aggregation across degree blocks; omitting one required fact may lower this substring-based score.

5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 3 reports the semantic-neighbor experiment across 11–51 tools ($N=50$, 3 counterbalanced tool-order blocks, 1,800 primary decisions).

Table 3. Scenario 3: Semantic-neighbor tool ambiguity ($N = 50$, 3 counterbalanced tool-order blocks).

Reg.	Config.	Vis.	Gold-R	Sel.	E2E	Tok	Lat
11	Full-reg single	11	—	94.0	86.0	1663	1098
	Top-10 single	10	100.0	93.3	83.3	1541	1148
	Routed gen.	5.1	98.0	90.7	80.0	1204	1889
	Routed spec.	5.1	98.0	93.3	85.3	1441	1983
31	Full-reg single	31	—	91.3	84.7	2706	1157
	Top-10 single	10	98.0	90.0	81.3	1149	1033
	Routed gen.	9.8	98.0	88.0	79.3	1396	1958
	Routed spec.	9.8	98.0	93.3	88.7	1830	2015
51	Full-reg single	51	—	92.7	83.3	3722	1289
	Top-10 single	10	96.0	86.7	78.7	1036	997
	Routed gen.	9.8	98.0	91.3	82.7	1340	1907
	Routed spec.	9.8	98.0	92.7	88.0	1770	2021

As the registry scales from 11 to 51 tools, Full-Registry Single Agent E2E success drops from 86.0% to 83.3% (input tokens +124%). Global Top-10 drops from 83.3% to 78.7% E2E (Gold Recall: 100.0% to 96.0%). Routed Specialist maintains 88.0% E2E at 51 tools (9.8 visible tools), yielding a +9.3 pp margin over Top-10 (95% CI $[+2.0, +17.3]$ pp) and +4.7 pp over Full-Registry (95% CI $[-1.3, +11.3]$ pp) with 52.4% fewer input tokens (1,770 vs. 3,722). Degradation relative to Top-10 is +7.3 pp (95% CI $[0.0, +15.3]$ pp).

Across 1,800 primary decisions, Routed Specialist latency ranges from 1,983 to 2,021 ms (supervisor routing: ~ 950 ms; input tokens: 1,441–1,830). Full-Registry latency rises from 1,098 to 1,289 ms as input tokens scale from 1,663 to 3,722.

5.4 Supporting Retrieval Validation

Table 4 presents a standalone offline evaluation of retrieval ($N=100$), isolating evidence retrieval performance from downstream multi-agent generation and routing. E4 cross-encoder reranks Hybrid RRF candidates; E5 reranks governed, hybrid, and BM25 candidates, then merges graph evidence under quotas.

Table 4. Standalone retrieval evaluation ($N=100$) on unique source filenames. E4: Hybrid RRF with cross-encoder reranking. E5: domain-governed, hybrid, and BM25 candidates with reranking and graph quotas. Evaluated independently of agent orchestration.

Configuration	Hit@1	Hit@3	P@5	R@5	MRR@10
E1: BM25 Lexical	0.530	0.770	0.206	0.617	0.669
E2: Dense Semantic	0.460	0.650	0.176	0.524	0.560
E3: Hybrid RRF	0.520	0.790	0.224	0.662	0.673
E4: Hybrid + Reranker	0.700	0.910	0.254	0.751	0.805
E5: Full Retrieval	0.740	0.920	0.302	0.861	0.833

BM25 outperforms dense search on Hit@1 (0.530 vs. 0.460), while Hybrid RRF balances both (0.790 Hit@3). The cross-encoder reranker yields the largest individual gain (+18.0 pp Hit@1). The full configuration reaches 0.740 Hit@1, 0.861 Recall@5, and 0.833 MRR@10; because it incorporates domain-specific components, the gain cannot be attributed to one component alone.

5.5 Failure Analysis

Table 5 categorizes errors in 58 failed query–configuration cases with Fact Coverage below 50% (34 for Single Agent and 24 for CTU-Chat). Labels were assigned through manual review of execution traces and intermediate outputs; categories are non-mutually exclusive.

Table 5. Failure taxonomy across architecture configurations ($N = 100$). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

Category	Failure Mode	S1-A	S1-C	Category	Failure Mode	S1-A	S1-C
Routing	Corrected misroute	N/A	1	Tool	Invalid arguments	4	1
	Repair misroute	N/A	1		Wrong-family call	3	1
Retrieval	Gold source absent	3	3	Generation	Unnecessary call	5	1
	Cross-domain omission	3	2		Result ignored	2	1
	Reranker demoted	7	6		Partial fact omission	22	14
Total Failed Queries (< 50% Fact):				S1-A (Single): 34 S1-C (CTU): 24			

The 34 Single Agent and 24 CTU-Chat failed query cases include 12 versus 3 tool errors. Retrieval errors are similar under shared evidence: 3 absent gold sources in each

arm and 7 versus 6 reranker demotions. With complete retrieval ($sr = 1.0$), partial fact omissions number 22 versus 14. These counts do not isolate effects of tool scoping or specialist prompts.

Three traces illustrate distinct stages of failure. For the multi-hop query HOUT-MHOP-ACAD-04, repair changed the correct `general` route to `academic`; that worker refused the request. For HOUT-XDOM-01, all gold sources were retrieved ($sr=1.0$), but the worker omitted the financial answer. For HOUT-XDOM-16, the query reached `scholarship`, but retrieval missed the target document ($sr=0.5$). Only the first is a route-repair failure; both cross-domain cases fail after owner selection.

6 Discussion

6.1 Architectural Findings and Operational Trade-offs

RQ1 (Architecture-Level Effectiveness): The tested orchestration bundle has a higher Automated Fact Pass Rate point estimate and fewer recorded specialist input tokens than Single Agent. The Fact Coverage confidence interval spans zero, and latency measures different pipeline stages; neither a completeness nor a speed advantage follows. Because the 50% threshold binarizes coverage, this margin does not establish more complete answers. The No Specialist Policy ablation is consistent with a contribution from domain instructions under the evaluated conditions.

RQ2 & RQ3 (Mechanisms, Tool Isolation, and Registry Scaling): The ablations associate specialist policies and heterogeneous evidence access with larger pass-rate losses when removed. Disabling deterministic calculators yields an estimated -4.0 pp point change in overall automated fact pass (95% CI $[-9.0, 0.0]$ pp); without route repair, cross-domain pass falls from 80.0% to 65.0%. At 11 tools, removing isolation increases input tokens without changing the pass estimate. At 51 tools, Routed Specialist records 88.0% tool E2E Success, 9.3 pp above Global Top-10 (CI $[+2.0, +17.3]$ pp); its 4.7 pp difference over Full-Registry Single Agent remains uncertain (CI $[-1.3, +11.3]$ pp). These results suggest a benefit from bounded tool exposure under the tested ambiguity.

Operational Trade-offs and Deployment Guidance: Single-agent timing begins after retrieval, whereas multi-agent figures include routing; these values cannot rank architectures by speed. Specialist execution records 1,731 ms and 677 output tokens more than generic execution. A single agent may suit narrow tasks with small tool pools; scoped specialists suit institutions with distinct evidence formats and fee calculations. Adapting CTU-Chat elsewhere requires mapping curriculum schemas, loading policies into Neo4j, PostgreSQL, and Qdrant, encoding fee rules in tools, updating intent labels and prompts, and testing against local regulations.

6.2 Relation to Prior Work

ReAct [18] and Toolformer [14] study tool use; AutoGen [16] supports agent coordination. CTU-Chat instead routes to bounded specialists with evidence paths chosen by data modality, differing from Adaptive-RAG’s complexity-based retrieval [4]. The Source Recall drop under Uniform Text-RAG aligns with GraphRAG’s rationale for relational evidence [2]. These frameworks were not directly compared in our experiments.

6.3 Threats to Validity

Internal and operational: The 10.65 s pipeline latency limits interactive use. Because Single Agent timing begins post-retrieval, measurements cannot rank cross-architecture speed; network variation adds uncertainty. **External:** Queries cover one institution; Scenario 3 uses synthetic tool neighbors. Transfer to other institutions and fee rules remains untested. Excluding Redis history leaves multi-turn effects unmeasured. **Construct:** Substring matching can pass an incorrect financial answer: HOUT-DIR-FIN-03 reached 50% coverage despite a 192.183M versus 40M VND discrepancy. Tool E2E checks verify execution validity rather than numeric exact match. **Statistical:** The 100-query paired analysis uses 10,000 bootstrap resamples; query family subsets have smaller sample sizes ($n = 19\text{--}33$).

7 Conclusion and Future Work

We evaluated CTU-Chat for university counseling through architecture comparisons, ablations, and tool ambiguity tests. Automated Fact Pass Rate exceeds the matched Single Agent by 10.0 pp (paired 95% CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$), with fewer specialist input tokens. Removing specialist policies or graph evidence lowers pass rate or Source Recall, respectively. At 51 tools, Routed Specialist records 88.0% tool E2E Success. Deterministic calculation ablation yields an estimated -4.0 pp difference in overall fact pass (95% CI $[-9.0, 0.0]$ pp); scoped access also bounds visible tools at 51-tool scale. To address offline fact-matching limitations, future work aims to combine advisor reviews, LLM-as-a-judge frameworks, and deterministic calculations. We also plan token streaming, caching, and prompt refinement to mitigate pipeline latency. Finally, student pilot trials will collect live colloquial queries for advisor annotation to expand the benchmark, while evaluating institutional portability.

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References

1. Cormack, G.V., Clarke, C.L.A., Buettcher, S.: Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 758–759 (2009). <https://doi.org/10.1145/1571941.1572114>
2. Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R.O., Larson, J.: From local to global: A graph RAG approach to query-focused summarization. arXiv preprint arXiv:2404.16130 (2024)
3. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, Q., Wang, H.: Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997 (2023)
4. Jeong, S., Baek, J., Cho, S., Hwang, S.J., Park, J.C.: Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 7036–7050 (2024). <https://doi.org/10.18653/v1/2024.naacl-long.389>

5. Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.-t.: Dense passage retrieval for open-domain question answering. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. pp. 6769–6781 (2020). <https://doi.org/10.18653/v1/2020.emnlp-main.550>
6. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems* **33**, 9459–9474 (2020)
7. Ma, T., La, T.T., Le Huu, L.T., Nguyen, M.N., Pham Luu, K.V.: REBot: From RAG to CatRAG with semantic enrichment and graph routing. In: *Multi-disciplinary Trends in Artificial Intelligence (MIWAI 2025)*. *Lecture Notes in Computer Science*, vol. 16354, pp. 435–447. Springer (2025). https://doi.org/10.1007/978-981-95-4960-3_35
8. Ma, T.T., Chau, T.K., Nguyen, T.P., Huynh, G.K., Nguyen, T.H., Tran, V.C., Do, T.N.: CAAS: Supporting admission counseling with Chat-Voice using ChatGPT and machine learning. In: *Proceedings of the 16th National Conference on Fundamental and Applied Information Technology Research (FAIR)*. pp. 526–533 (2023). <https://doi.org/10.15625/vap.2023.0079>
9. Nguyen, D.Q., Nguyen, A.T.: PhoBERT: Pre-trained language models for Vietnamese. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*. pp. 1037–1042 (2020). <https://doi.org/10.18653/v1/2020.findings-emnlp.92>
10. Nogueira, R., Cho, K.: Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085* (2019)
11. Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., Wu, X.: Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering* **36**(7), 3580–3599 (2024). <https://doi.org/10.1109/TKDE.2024.3352100>
12. Qian, J., Wang, H., Li, Z., Li, S., Yan, X.: Limitations of language models in arithmetic and symbolic induction. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. pp. 9285–9298 (2023). <https://doi.org/10.18653/v1/2023.acl-long.516>
13. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval* **3**(4), 333–389 (2009). <https://doi.org/10.1561/15000000019>
14. Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Hambro, E., Zettlemoyer, L., Cancedda, N., Scialom, T.: Toolformer: Language models can teach themselves to use tools. In: *Advances in Neural Information Processing Systems*. vol. 36 (2023). <https://doi.org/10.52202/075280-2997>
15. Wollny, S., Schneider, J., Di Mitri, D., Weidlich, J., Rittberger, M., Drachsler, H.: Are we there yet? a systematic literature review on chatbots in education. *Frontiers in Artificial Intelligence* **4**, 654924 (2021). <https://doi.org/10.3389/frai.2021.654924>
16. Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., et al.: AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155* (2023)
17. Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., Nie, J.Y.: C-Pack: Packed resources for general Chinese embeddings. In: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. pp. 641–649 (2024). <https://doi.org/10.1145/3626772.3657878>
18. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Synergizing reasoning and acting in language models. In: *International Conference on Learning Representations (ICLR)* (2023)